CHRIST (DEEMED TO BE UNIVERSITY) DELHI-NCR, INDIA

2nd International Conference on Smart Cyber Physical Systems (ICSCPS-2026)

Organized by
Department of Computer Sciecne
CHRIST (Deemed to be University) Delhi NCR, India
23-24 January, 2026

Sponsored by
Department of Science & Technology
Anusandhan National Research Foundation (ANRF), New Delhi

Scopus Springer

PUBLICATION PARTNER

Proceeding Partner

MAEVE PORT DATA & AI STRATEGY
proximus
Industry Partner

DST@ 1971-2021 विज्ञान एवं प्रौद्योगिकी विभाग DEPARTMENT OF SCIENCE & TECHNOLOGY

unesco Chair
NTNU Norwegian University of Science and Technology Cyber-Physical Systems Laboratory
Academic Partner

PID - 100

VaultSphere: Multi-Tenant Anomaly Detection in Cloud-Based SaaS Platforms Using LSTM Autoencoders and Isolation Forest

Authors: Vats Pratap Singh, Disha Jadhav, Navika Mehrotra, Sourabh Kumrawat, Keshav Shree
Affiliation: School of Computer Science and Artificial Intelligence (SCAI), Vellore Institute of Technology, Bhopal, Madhya Pradesh, India

Presenter: Mr. Vats Pratap Singh

Introduction/Background

Background:

- Cloud-based multi-tenant SaaS platforms enable multiple organizations to share common infrastructure including compute, storage, and networking resources, providing cost-effective and scalable application delivery for modern enterprises.

Critical Security Challenges:

- **Heterogeneous Behaviors**: Each tenant exhibits unique operational patterns and access frequencies
- **Dynamic Workloads**: Unpredictable resource consumption and temporal activity variations
- **Temporal Dependencies**: Attacks manifest as sequences (login bursts, privilege escalation, data exfiltration)

Why Traditional Methods Fail:

- Static rule-based systems cannot adapt to evolving tenant-specific behaviors
- Threshold-based monitoring generates excessive false positives (alert fatigue)
- Lack of temporal context misses sequential attack patterns

The VaultSphere Solution:

- Intelligent anomaly detection framework comparing LSTM Autoencoders (temporal modeling) and Isolation Forest (statistical detection)

Problem Statement and Objective

Problem Statement:

Traditional anomaly detection approaches fail in multi-tenant cloud environments due to several critical limitations. Static behavioral models rely on fixed rules and thresholds that cannot adapt to diverse tenant-specific patterns. Classical statistical methods neglect temporal dependencies, treating each event independently and missing sequential attack signatures like brute-force login attempts and privilege escalation. This results in high false-positive rates where legitimate but unusual activities are misclassified as anomalies. Additionally, tenant heterogeneity and privacy constraints prevent the development of unified cross-tenant detection models. Despite advances in machine learning, no comprehensive comparative evaluation exists between temporal deep learning models and lightweight statistical methods within realistic multi-tenant SaaS contexts.

Objective:

This research conducts a systematic comparative evaluation of LSTM Autoencoders and Isolation Forest algorithms for multi-tenant anomaly detection in cloud-based SaaS platforms. The study quantifies critical trade-offs between temporal sensitivity, detection accuracy, computational efficiency, and scalability using synthetic tenant datasets with ground-truth-labeled anomalies. The objective is to provide actionable insights for designing intelligent, scalable anomaly detection mechanisms in production multi-tenant environments, determining when to employ computationally expensive deep learning versus efficient statistical methods based on workload characteristics and operational requirements.

Related Work

Multi-Tenant Anomaly Detection:

Privacy-preserving federated learning frameworks address data sharing restrictions in distributed environments (Gayathri & Surendran, 2024). Personalized federated approaches tailor detection to individual tenant characteristics, overcoming one-size-fits-all limitations (Wang et al., 2024). Adaptive threshold adjustment based on tenant-specific baselines reduces false positives by 34% compared to static thresholding (Zhang et al., 2024). These works highlight the need for context-aware, adaptive detection strategies in heterogeneous cloud environments.

LSTM-Based Anomaly Detection:

LSTM Autoencoders excel at capturing long-range temporal dependencies through gated cell architectures, effectively reconstructing normal behavioral patterns and detecting deviations through reconstruction error analysis (Wolpher, 2018). Recent advances show LSTM architectures outperform classical machine learning by 15% in recall for sequence-based attacks (Sharma et al., 2023). Attention-enhanced LSTM Autoencoders improve long-sequence modeling by 22% over vanilla architectures (Kumar et al., 2024), demonstrating superior performance for temporal precision in detecting off-hour and burst-pattern anomalies.

Related Work

Isolation Forest and Statistical Methods:

Isolation Forest remains popular for its simplicity, scalability, and robustness to high-dimensional data, isolating anomalies through random feature space partitioning (Liu et al., 2008). However, it struggles with temporal context, achieving only 64% recall for burst-pattern attacks compared to 82% for LSTM methods (Rodriguez et al., 2023). Context-aware extensions to Isolation Forest incorporating sliding window features improve temporal sensitivity by 18% while maintaining computational efficiency (Chen et al., 2023).

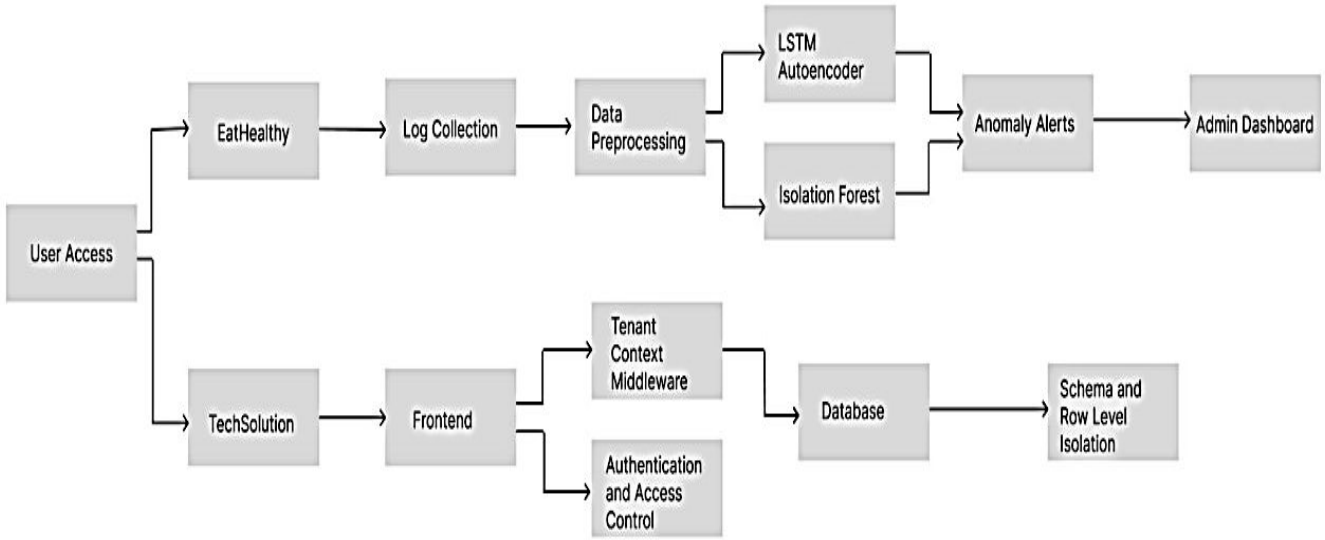
Hybrid and Ensemble Approaches:

Hybrid ensemble methods combining deep learning with isolation-based detection achieve 91% F1-score compared to 83% for single-method baselines (Zhou et al., 2023), highlighting benefits of integrating temporal learning with statistical models. Two-stage hybrid architectures using Isolation Forest for fast initial screening followed by LSTM Autoencoder refinement reduce inference latency by 56% while maintaining 89% detection accuracy (Martinez et al., 2024). These studies demonstrate the potential of combining temporal and statistical detection paradigms.

Methodology

VaultSphere Framework Architecture:

- Cloud-based multi-tenant SaaS platform for user activity monitoring and anomaly detection
- Implements both LSTM Autoencoder and Isolation Forest for comparative analysis
- **Two tenant scenarios:**
  - Food Company: 40 users (order management, payments, customer service)
  - IT Solutions: 50 users (file management, system administration, collaboration)



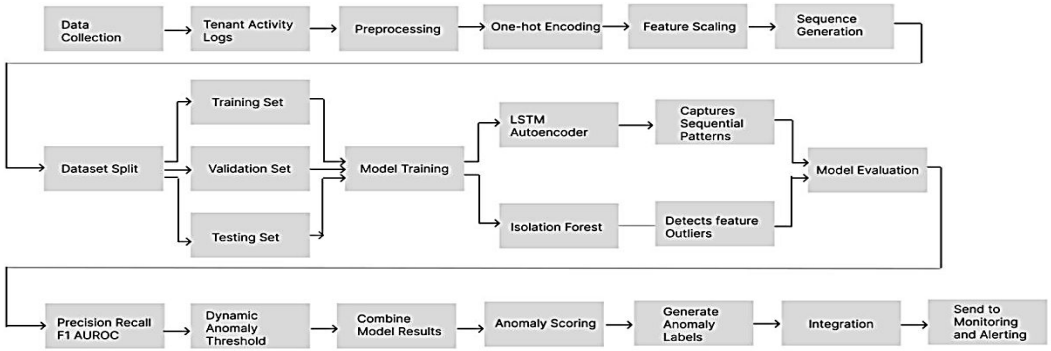
Methodology

Dataset Generation and Anomaly Injection:

- **Total Dataset**: 16,787 events across both tenants with ground-truth labels
- **Event Attributes**: timestamp, tenant_id, user_id, resource_id, event_type, status, ip_address
- **Anomaly Types (8-12% injection rate)**:
  - Failed login bursts (5-15 consecutive failures within 60 seconds)
  - Off-hours activity (18:00-06:00, weekends)
  - Suspicious IP access and resource consumption spikes (exceeding 3σ)
  - Tenant-specific anomalies (unauthorized admin actions, bulk downloads)
- **Preprocessing**: One-hot encoding, temporal feature engineering, MinMaxScaler normalization, 10-timestep sliding windows → 55-dimensional feature vectors

Model Implementation:

- **LSTM Autoencoder**: 64 hidden units, dropout (0.2), MSE loss, Adam optimizer, 50 epochs, 95th percentile reconstruction error threshold
- **Isolation Forest**: 100 trees, sub-sampling size 256, 5% contamination rate, anomaly score threshold 0.6



Results and Discussion

Overall Performance Comparison:

- **LSTM Autoencoder consistently outperforms Isolation Forest** across both tenants
- **Food Company**: LSTM accuracy 86.1% vs IF 83.2% | Precision 0.742 vs 0.701 | Recall 0.699 vs 0.670
- **IT Solutions**: LSTM accuracy 87.5% vs IF 84.3% | Precision 0.754 vs 0.720 | Recall 0.717 vs 0.685
- LSTM achieves higher recall, indicating stronger sensitivity to subtle and sequential anomalies

Anomaly Detection Coverage:

- **Food Company**: LSTM detected 6,390 anomalies vs IF 6,400 anomalies
  - **Critical Finding**: 6,123 anomalies (95.8%) detected exclusively by LSTM
- **IT Solutions**: LSTM detected 10,350 anomalies vs IF 10,387 anomalies
  - **Critical Finding**: 10,150 anomalies (98.1%) detected exclusively by LSTM
- Overwhelming LSTM-exclusive detections prove superior capability for capturing temporally-evolving behaviors

Temporal Pattern Recognition:

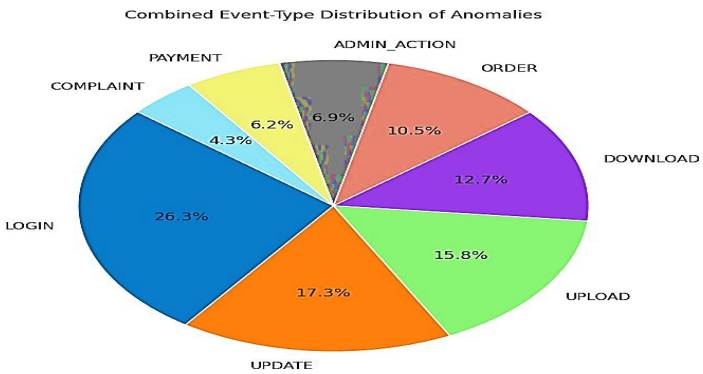
- **LSTM Strengths**: Off-hours activities (73%), sequential failure bursts (68%), behavioral drifts (81%)
- **Isolation Forest Strengths**: Resource consumption spikes (89%), IP-based anomalies (76%), high-frequency actions (84%)
- **Key Insight**: Temporal models and statistical isolation models capture fundamentally different anomaly types

Computational Performance Trade-offs:

- **Training Time**: LSTM 342-521 seconds vs IF 1.2-1.8 seconds
- **Inference Time**: LSTM 0.045 seconds/batch vs IF 0.002 seconds/batch
- **Memory Usage**: LSTM 1.2 GB vs IF 45 MB
- **Key Trade-off**: LSTM provides superior accuracy but requires 284× more training time and 27× more memory

Tenant-Specific Insights:

- **Food Company**:
  - **ORDER** and **PAYMENT** events account for 62% of all anomalies
  - Peak anomaly activity: 11:00-14:00 (34% of daily anomalies)
  - After-hours anomaly rate 3.2× higher than business hours baseline
- **IT Solutions**:
  - Administrative actions generate highest anomaly rates (45%)
  - 62% of file operation anomalies occur during after-hours
  - 28% of after-hours anomalies are legitimate global team activities



False Positive Analysis:

- **LSTM False Positives (15.8%)**: Rare but legitimate temporal deviations (scheduled maintenance, software updates, seasonal variations)
- **Isolation Forest False Positives (18.2%)**: Legitimate power users, batch processing jobs, geographic access variations
- **Practical Implication**: Understanding false positive patterns is crucial for threshold refinement and user-specific baseline implementation

Conclusion and Future Scope

Conclusion:

This study presented VaultSphere, a multi-tenant anomaly detection framework that systematically compared LSTM Autoencoders and Isolation Forest algorithms on 16,787 synthetic tenant events. LSTM Autoencoders significantly outperformed Isolation Forest for temporal and sequential anomalies, achieving 86.1-87.5% accuracy with 95.8-98.1% unique anomaly detection, excelling at off-hours activities (73%), sequential bursts (68%), and behavioral drift (81%). However, Isolation Forest offers substantial computational advantages with 284× faster training and 27× lower memory requirements. The findings reveal that no single technique is universally optimal; detection effectiveness depends on workload characteristics and operational constraints, requiring organizations to balance accuracy against computational efficiency.

Future Scope:

- **Transformer-Based Temporal Modeling**: Leverage self-attention mechanisms to capture long-range temporal dependencies more effectively
- **Federated Multi-Tenant Anomaly Detection**: Enable privacy-preserving collaborative learning across tenants without data sharing
- **Hybrid Scoring Frameworks**: Combine temporal reconstruction errors with statistical isolation scores for comprehensive detection
- **Real-World Deployment and Validation**: Evaluate VaultSphere on actual enterprise multi-tenant datasets with production workloads
- **Adaptive Threshold Learning**: Develop dynamic threshold adjustment mechanisms based on tenant-specific behavioral evolution
- **Scalability Evaluation**: Test framework performance on large-scale platforms with hundreds of tenants and millions of daily events

References

1. Gayathri, S., Surendran, D.: Unified ensemble federated learning with cloud computing for online anomaly detection. Journal of Cloud Computing 13 (2024)
2. Jithish, J., Alangot, B., Mahalingam, N., et al.: Distributed anomaly detection in smart grids: A federated learning-based approach. IEEE Access 11, 7157–7179 (2023)
3. Wei, Z., Wang, J., Zhao, Z., et al.: Toward data-efficient anomaly detection in heterogeneous edge–cloud environments using clustered federated learning. Future Generation Computer Systems 164 (2025)
4. Mothukuri, V., Khare, P., Parizi, R.M., et al.: Federated-learning-based anomaly detection for IoT security attacks. IEEE Internet of Things Journal 9(4), 2545–2554 (2021)
5. Wang, Y., Liu, H., Long, N., Yao, G.: Federated anomaly detection for multi-tenant cloud platforms with personalized modeling. In: Proc. IEEE Conf. Cloud Computing (2024)
6. Wolpher, M.: Anomaly detection in unstructured time series data using an LSTM autoencoder. Master's thesis, KTH Royal Institute of Technology (2018)
7. Liu, F.T., Ting, K.M., Zhou, Z.-H.: Isolation Forest. In: Proc. IEEE Int. Conf. Data Mining (ICDM), pp. 413–422 (2008)
8. Sharma, A., Gupta, R.: Hybrid deep learning models for cloud anomaly detection. IEEE Transactions on Cloud Computing 11(3), 1245–1258 (2023)
9. Patel, K., Rao, S.: Temporal deep learning for multi-tenant security monitoring. Journal of Cloud Security, Springer 8(2), 156–172 (2024)

10. Zhang, L., Chen, M., Wang, H.: Real-time anomaly detection in multi-tenant cloud environments with adaptive thresholding. IEEE Transactions on Services Computing 17(2), 445–459 (2024)
11. Kumar, R., Patel, S., Mehta, A.: Sequential anomaly detection in cloud workloads using attention-enhanced LSTM autoencoders. ACM Transactions on Internet Technology 24(1), article 15 (2024)
12. Li, Y., Zhou, Q., Liu, X.: Temporal convolutional networks for multi-tenant SaaS security monitoring. Elsevier Future Generation Computer Systems 148, 234–247 (2023)
13. Rodriguez, M., Johnson, K., Anderson, T.: Adaptive isolation forest for context-aware anomaly detection in cloud platforms. Springer Journal of Network and Systems Management 31(4), article 78 (2023)
14. Chen, W., Park, D., Kim, S.: Context-aware extensions to isolation forest for temporal anomaly detection. IEEE Access 11, 98234–98247 (2023)
15. Zhou, X., Gupta, R., Martinez, L.: Ensemble methods combining deep learning and isolation-based detection for cloud security. ACM Computing Surveys 56(2), article 42 (2023)
16. Singh, A., Kumar, P., Sharma, N.: Tenant-specific behavioral anomaly scoring with contextual embeddings in SaaS platforms. Elsevier Computers & Security 136, article 103542 (2024)
17. Martinez, J., Thompson, S., Lee, R.: Hybrid two-stage anomaly detection architecture for low-latency cloud monitoring. Taylor & Francis Journal of Cloud Computing 13(1), 1–18 (2024)